

培道中學校

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年級：

科目：

練習題 15.1

2. 考慮在 (x_0, y_0, z_0) 處的切平面： $2x_0(x-x_0)+4y_0(y-y_0)+6z_0(z-z_0)=0$

$$\Leftrightarrow x_0x+2y_0y+3z_0z=21$$

因為切平面與平面 $x+4y+6z=0$ 平行，所以 $\exists k \in \mathbb{R}$ ， $\begin{pmatrix} x_0 \\ 2y_0 \\ 3z_0 \end{pmatrix} = \begin{pmatrix} -k \\ 4k \\ 6k \end{pmatrix}$

$$\Rightarrow 21 = x_0^2 + 2y_0^2 + 3z_0^2 = k^2 + (2k)^2 + (2k)^2 = 9k^2$$

$$k = \pm \frac{\sqrt{21}}{3}$$

因此，所求切平面是 $\frac{\sqrt{21}}{3}x + \frac{4\sqrt{21}}{3}y + \frac{6\sqrt{21}}{3}z - 21 = 0$ 和 $-\frac{\sqrt{21}}{3}x - \frac{4\sqrt{21}}{3}y - \frac{6\sqrt{21}}{3}z + 21 = 0$

即 $x+4y+6z=3\sqrt{21}$ 和 $x+4y+6z=-3\sqrt{21}$

4. 設公共切點是 $P(x_0, y_0, z_0)$

對於曲面 $xyz = \lambda$ ，其切平面（在 P 點）是 $y_0z_0(x-x_0)+z_0x_0(y-y_0)+x_0y_0(z-z_0)=0$

$$\Leftrightarrow y_0z_0x+z_0x_0y+x_0y_0z=3\lambda$$

對於橢球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ ，其切平面（在 P 點）是 $\frac{2x_0}{a^2}(x-x_0)+\frac{2y_0}{b^2}(y-y_0)+\frac{2z_0}{c^2}(z-z_0)=0$

$$\Leftrightarrow \frac{x_0}{a^2}x+\frac{y_0}{b^2}y+\frac{z_0}{c^2}z=1$$

因為 $\lambda > 0$ ，所以 $x, y, z \neq 0$

$$\text{因此 } \frac{a^2y_0z_0}{x_0} = \frac{b^2z_0x_0}{y_0} = \frac{c^2x_0y_0}{z_0} = 3\lambda \Rightarrow (3\lambda)^3 = a^2b^2c^2x_0y_0z_0 = a^2b^2c^2\lambda$$

$$\Leftrightarrow \lambda^2 = \frac{1}{27}a^2b^2c^2 \Rightarrow \lambda = \frac{\sqrt{3}}{9}|abc| = \frac{\sqrt{3}}{9}abc$$

5. 取兩者任一交點 (x_0, y_0, z_0) 。對於圓柱面 $x^2+y^2=a^2$ ，在 (x_0, y_0, z_0) 處的法向量是 $\vec{n}_1 = \begin{pmatrix} x_0 \\ y_0 \\ 0 \end{pmatrix}$
對於曲面 $bz=xy$ ，在 (x_0, y_0, z_0) 處的法向量是 $(y_0, x_0, -b) = \vec{n}_2$

$$\begin{aligned} \cos \angle(\vec{n}_1, \vec{n}_2) &= \frac{\vec{n}_1 \cdot \vec{n}_2}{\|\vec{n}_1\| \|\vec{n}_2\|} = \frac{2x_0y_0}{\sqrt{x_0^2+y_0^2} \cdot \sqrt{x_0^2+y_0^2+b^2}} \Leftrightarrow \angle(\vec{n}_1, \vec{n}_2) = \arccos \left(\frac{2x_0y_0}{\sqrt{x_0^2+y_0^2} \cdot \sqrt{x_0^2+y_0^2+b^2}} \right) \\ &= \arccos \left(\frac{2bz_0}{a\sqrt{a^2+b^2}} \right) \end{aligned}$$

6. 在 (x_0, y_0, z_0) 處，橢球面的法向量是 $(\frac{x_0}{a^2}, \frac{y_0}{b^2}, \frac{z_0}{c^2})$

$$\text{由題意可得 } \langle (\frac{x_0}{a^2}, \frac{y_0}{b^2}, \frac{z_0}{c^2}), (1, 0, 0) \rangle = \langle (\frac{x_0}{a^2}, \frac{y_0}{b^2}, \frac{z_0}{c^2}), (0, 0, 1) \rangle$$

$$\Leftrightarrow \frac{x_0}{a^2} = \frac{z_0}{c^2}$$

$$\text{設 } x_0 = ka^2, z_0 = kc^2, \text{ 那麼 } y_0^2 = b^2 [1 - k^2(a^2 + c^2)]$$

$$\Leftrightarrow y_0 = \pm b \sqrt{1 - k^2(a^2 + c^2)}$$

$$\text{因此，所求點集是 } \{(x, y, z) \mid x = ka^2, z = kc^2, y = \pm b \sqrt{1 - k^2(a^2 + c^2)}, \forall k \leq \frac{1}{a^2 + c^2}\}$$

7. 在 (x_0, y_0, z_0) 處，曲面 $x^2 - x + y^2 + z^2 = 0$ 的法向量是 $(2x_0 - 1, 2y_0, 2z_0) = \vec{n}_1$

平面 $x - y - z = 2$ 的法向量是 $(1, -1, -1) = \vec{n}_2$

平面 $x - y - \frac{z}{2} = 2$ 的法向量是 $(1, -1, -\frac{1}{2}) = \vec{n}_3$

因為 $\vec{n}_1 \perp \vec{n}_2, \vec{n}_1 \perp \vec{n}_3$ ，所以 $\vec{n}_1 \parallel \vec{n}_2 \times \vec{n}_3$

$$\vec{n}_2 \times \vec{n}_3 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & -1 \\ 1 & -1 & -\frac{1}{2} \end{vmatrix} = (-\frac{1}{2}, -\frac{1}{2}, 0)$$

$$\text{所以 } \exists k \in \mathbb{R}, \text{ 使得 } (2x_0 - 1, 2y_0, 2z_0) = (-\frac{k}{2}, -\frac{k}{2}, 0)$$

$$\Rightarrow z_0 = 0 \text{ 且 } x_0 = \frac{1 - \frac{k}{2}}{2}, y_0 = -\frac{k}{4}$$

$$0 = (x_0 - 1)x_0 + y_0^2 + z_0^2 = -\frac{1 + \frac{k}{2}}{2} \cdot \frac{1 - \frac{k}{2}}{2} + \frac{k^2}{16} = \frac{1}{4} \left(\frac{k^2}{4} - 1 \right) + \frac{k^2}{16} = \frac{k^2}{8} - \frac{1}{4}$$

$$\Rightarrow k = \pm \sqrt{2}$$

$$(i) \text{ 當 } k = \sqrt{2} \text{ 時, } (x_0, y_0, z_0) = (\frac{2 - \sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, 0)$$

對應的切平面是

$$-\frac{\sqrt{2}}{2} \left(x - \frac{2 - \sqrt{2}}{4} \right) - \frac{\sqrt{2}}{2} \left(y + \frac{\sqrt{2}}{4} \right) = 0 \Leftrightarrow x + y = \frac{1 - \sqrt{2}}{2}$$

$$(ii) \text{ 當 } k = -\sqrt{2} \text{ 時, } (x_0, y_0, z_0) = (\frac{2 + \sqrt{2}}{4}, \frac{\sqrt{2}}{4}, 0)$$

對應的切平面是

$$\frac{\sqrt{2}}{2} \left(x - \frac{2 + \sqrt{2}}{4} \right) + \frac{\sqrt{2}}{2} \left(y - \frac{\sqrt{2}}{4} \right) = 0 \Leftrightarrow x + y = \frac{1 + \sqrt{2}}{2}$$

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8. 在 (x_0, y_0, z_0) 處的切平面是 $\frac{1}{\sqrt{x_0}}(x-x_0) + \frac{1}{\sqrt{y_0}}(y-y_0) + \frac{1}{\sqrt{z_0}}(z-z_0) = 0$

$$\Leftrightarrow \frac{x}{\sqrt{x_0}} + \frac{y}{\sqrt{y_0}} + \frac{z}{\sqrt{z_0}} = \sqrt{a}$$

在 x 軸上的截距是 $x \Big|_{y=0, z=0} = \sqrt{a}\sqrt{x_0}$

同理可得 y 軸上和 z 軸上截距分別是 $\sqrt{a}\sqrt{y_0}$ 和 $\sqrt{a}\sqrt{z_0}$

所以，割下線段之和是 $|\sqrt{a}\sqrt{x_0}| + |\sqrt{a}\sqrt{y_0}| + |\sqrt{a}\sqrt{z_0}| = \sqrt{a}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}) = (\sqrt{a})^2 = a$ 為常數

9. 設 $u(x, y, z) = F(x-az, y-bz)$, $\varepsilon = x-az$, $\eta = y-bz$

在 (x_0, y_0, z_0) 處的法向量是 $(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}) = (\frac{\partial F}{\partial \varepsilon}, \frac{\partial F}{\partial \eta}, -a\frac{\partial F}{\partial \varepsilon} - b\frac{\partial F}{\partial \eta}) = \vec{n}$

考慮直線 $L: \{(ta, tb, t) \mid t \in \mathbb{R}\}$, L 的方向向量 $\vec{e} = (a, b, 1)$

$$\langle \vec{n}, \vec{e} \rangle = a\frac{\partial F}{\partial \varepsilon} + b\frac{\partial F}{\partial \eta} - a\frac{\partial F}{\partial \varepsilon} - b\frac{\partial F}{\partial \eta} = 0 \Leftrightarrow \vec{n} \perp \vec{e}$$

$\Leftrightarrow$ 任一切平面都與直線 L 平行

10. 記 $u(x, y, z) = F(\frac{x-a}{z-c}, \frac{y-b}{z-c})$, $\varepsilon = \frac{x-a}{z-c}$, $\eta = \frac{y-b}{z-c}$

對於任一點 $P(x_0, y_0, z_0)$, 其上的切平面是 $\frac{\partial F}{\partial \varepsilon} \cdot \frac{1}{z_0-c} (x-x_0) + \frac{\partial F}{\partial \eta} \cdot \frac{1}{z_0-c} (y-y_0) + (\frac{\partial F}{\partial \varepsilon} \cdot (-\frac{x_0-a}{(z_0-c)^2}) + \frac{\partial F}{\partial \eta} \cdot (-\frac{y_0-b}{(z_0-c)^2})) (z-z_0) = 0$

$$\text{整理後得 } \frac{\partial F}{\partial \varepsilon} \cdot (x-x_0) + \frac{\partial F}{\partial \eta} \cdot (y-y_0) = \left[\frac{\partial F}{\partial \varepsilon} \cdot \left(\frac{x_0-a}{z_0-c}\right) + \frac{\partial F}{\partial \eta} \cdot \left(\frac{y_0-b}{z_0-c}\right) \right] (z-z_0) \dots\dots (1)$$

注意到, 代入 $(x, y, z) = (a, b, c)$ 到方程(1)

$$\text{左式} = -\frac{\partial F}{\partial \varepsilon} (x_0-a) - \frac{\partial F}{\partial \eta} (y_0-b), \text{右式} = \left[\frac{\partial F}{\partial \varepsilon} \left(\frac{x_0-a}{z_0-c}\right) + \frac{\partial F}{\partial \eta} \left(\frac{y_0-b}{z_0-c}\right) \right] (c-z_0)$$

$$= -\frac{\partial F}{\partial \varepsilon} (x_0-a) - \frac{\partial F}{\partial \eta} (y_0-b) = \text{左式}$$

所以 (a, b, c) 是曲面上任一切平面內的點。

11. 在 $\vec{p}_0 = (x_0, y_0, z_0)$ 處上, 曲面的法向量是 (記 $r = x^2 + y^2 + z^2$)

$$\vec{n} = \left(\frac{\partial \Phi}{\partial r} \cdot 2x_0 - a, \frac{\partial \Phi}{\partial r} \cdot 2y_0 - b, \frac{\partial \Phi}{\partial r} \cdot 2z_0 - c \right) = 2 \frac{\partial \Phi}{\partial r} \vec{p}_0 - (a, b, c)$$

因此 $\vec{n}$ 可以用 $\vec{p}_0, (a, b, c)$ 線性表出 $\Leftrightarrow \vec{n}, \vec{p}_0, (a, b, c)$ 共面

12. 任取交點集內的一點 (x_0, y_0, z_0) .

對於曲面 $F(x, y, z) = 0$, 其上的切平面是 $\frac{\partial F}{\partial x}(x - x_0) + \frac{\partial F}{\partial y}(y - y_0) + \frac{\partial F}{\partial z}(z - z_0) = 0$

對於曲面 $G(x, y, z) = 0$, 其上的切平面是 $\frac{\partial G}{\partial x}(x - x_0) + \frac{\partial G}{\partial y}(y - y_0) + \frac{\partial G}{\partial z}(z - z_0) = 0$

$$\text{則所求切線是 } \left(\frac{\partial F}{\partial x} \frac{\partial G}{\partial z} - \frac{\partial F}{\partial z} \frac{\partial G}{\partial x} \right) (x - x_0) + \left(\frac{\partial F}{\partial y} \frac{\partial G}{\partial z} - \frac{\partial F}{\partial z} \frac{\partial G}{\partial y} \right) (y - y_0) = 0$$

$$13. \begin{cases} x^2 + y^2 + z^2 = 9 \\ x^2 - y^2 = 3 \end{cases} \Leftrightarrow \begin{cases} x^2 - y^2 = 3 \\ 2y^2 + z^2 = 6 \end{cases} \Leftrightarrow \begin{cases} 2x^2 + z^2 = 12 \\ 2y^2 + z^2 = 6 \end{cases}$$

顯然地, 由連續性可知在 $(2, 1, 2)$ 附近有 $x = \sqrt{6 - \frac{z^2}{2}}, y = \sqrt{3 - \frac{z^2}{2}}$

記 $\vec{r}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$ 由上述條件知在 $(2, 1, 2)$ 附近有 $\vec{r}(t) = \begin{pmatrix} \sqrt{6 - \frac{t^2}{2}} \\ \sqrt{3 - \frac{t^2}{2}} \\ t \end{pmatrix}$

$$\text{曲率 } k(t) = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|^3}, \quad \vec{r}'(t) = \begin{pmatrix} \frac{-t}{2\sqrt{6 - \frac{t^2}{2}}} \\ \frac{-t}{2\sqrt{3 - \frac{t^2}{2}}} \\ 1 \end{pmatrix}, \quad \vec{r}''(t) = \begin{pmatrix} \frac{-\frac{1}{2} \cdot \left(\sqrt{6 - \frac{t^2}{2}} - t \cdot \frac{-t}{2\sqrt{6 - \frac{t^2}{2}}} \right)}{6 - \frac{t^2}{2}} \\ \frac{-\frac{1}{2} \cdot \left(\sqrt{3 - \frac{t^2}{2}} - t \cdot \frac{-t}{2\sqrt{3 - \frac{t^2}{2}}} \right)}{3 - \frac{t^2}{2}} \\ 0 \end{pmatrix}$$

$$\Rightarrow \vec{r}'(2) = \begin{pmatrix} -\frac{1}{2} \\ -1 \\ 1 \end{pmatrix}, \quad \vec{r}''(2) = \begin{pmatrix} -\frac{3}{8} \\ -\frac{3}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{3}{(6 - \frac{t^2}{2})^{\frac{3}{2}}} \\ -\frac{3}{(3 - \frac{t^2}{2})^{\frac{3}{2}}} \\ 0 \end{pmatrix}$$

$$\vec{r}'(2) \times \vec{r}''(2) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\frac{1}{2} & -1 & 1 \\ -\frac{3}{8} & -\frac{3}{2} & 0 \end{vmatrix} = \left(\frac{3}{2}, -\frac{3}{8}, \frac{3}{8} \right)$$

$$k(2) = \frac{\sqrt{\left(\frac{3}{2}\right)^2 + \left(-\frac{3}{8}\right)^2 + \left(\frac{3}{8}\right)^2}}{\left(\sqrt{\left(-\frac{1}{2}\right)^2 + (-1)^2 + 1^2}\right)^3} = \frac{\frac{3}{8} \times 3\sqrt{2}}{\left(\frac{3}{2}\right)^3} = \frac{\sqrt{2}}{3}$$

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練習題 15.2

$$1. (1) \quad E = \|(a \cos u \cos v, b \cos u \sin v, -c \sin u)\|^2$$

$$= a^2 \cos^2 u \cos^2 v + b^2 \cos^2 u \sin^2 v + c^2 \sin^2 u$$

$$F = a \cos u \cos v \cdot (-a \sin u \sin v) + b \cos u \sin v \cdot (b \sin u \cos v)$$

$$= \frac{(b^2 - a^2)}{4} \sin 2u \sin 2v$$

$$G = \|(-a \sin u \sin v, b \sin u \cos v, 0)\|^2 = (a^2 \sin^2 v + b^2 \cos^2 v) \sin^2 u$$

(4)

$$E = a^2 + b^2 + 4v^2, \quad F = a^2 + b \cdot (-b) + (2v) \cdot (2u) = 4uv + a^2 - b^2$$

$$G = a^2 + b^2 + 4u^2$$

$$(6) \quad E = \cos^2 v + \sin^2 v + \left(\frac{d\varphi}{du}\right)^2, \quad F = \cos v \cdot (-u \sin v) + (\sin v) \cdot (u \cos v) + \frac{d\varphi}{du} \cdot a = a \frac{d\varphi}{du}$$

$$G = (-u \sin v)^2 + (u \cos v)^2 + a^2 = a^2 + u^2$$

2. 證：

$$\begin{aligned} d\vec{r}^2 &= \left(\frac{\partial \vec{r}}{\partial u} du + \frac{\partial \vec{r}}{\partial v} dv \right)^2 = \left(\frac{\partial \vec{r}}{\partial u} \right)^2 du^2 + 2 \left\langle \frac{\partial \vec{r}}{\partial u}, \frac{\partial \vec{r}}{\partial v} \right\rangle du dv + \left(\frac{\partial \vec{r}}{\partial v} \right)^2 dv^2 \\ &= E du^2 + 2F du dv + G dv^2 \end{aligned}$$

3. 證：

$$\vec{r} = \vec{r}(u, v) = \vec{R}(v)$$

$$d\vec{r} = \left(\frac{\partial \vec{r}}{\partial u} + \frac{\partial \vec{r}}{\partial v} \right) dv \Rightarrow ds = \sqrt{E + 2F + G} dv$$

$$\text{因為 } I = du^2 + (u^2 + a^2) dv^2, \text{ 所以 } E=1, F=0, G=u^2 + a^2$$

$$ds = \sqrt{1 + (u^2 + a^2)} dv = \sqrt{(v^2 + a^2) + 1} dv$$

$$S = \int_{v_1}^{v_2} ds = \int_{v_1}^{v_2} \sqrt{v^2 + a^2 + 1} dv$$

設 $v = \sqrt{a^2+1} \sinh t$, $dv = \sqrt{a^2+1} \cosh t dt$

$$S = \int_{t_1}^{t_2} (\sqrt{a^2+1} \cosh t) \cdot (\sqrt{a^2+1} \cosh t) dt = \int_{t_1}^{t_2} (a^2+1) \frac{\cosh(2t)+1}{2} dt$$

$$= \frac{a^2+1}{2} (\cosh(2t_2) - \cosh(2t_1) + t_2 - t_1)$$

$$v_1 = \sqrt{a^2+1} \sinh t_1 = \sqrt{a^2+1} \cdot \frac{e^{t_1} - e^{-t_1}}{2} \Leftrightarrow (e^{t_1})^2 - \frac{2v_1}{\sqrt{a^2+1}} e^{t_1} - 1 = 0$$

$$\Leftrightarrow \left(e^{t_1} - \frac{v_1}{\sqrt{a^2+1}}\right)^2 = \frac{v_1^2}{a^2+1} + 1 \Leftrightarrow e^{t_1} = \frac{v_1}{\sqrt{a^2+1}} + \sqrt{\frac{v_1^2}{a^2+1} + 1}$$

$$t_1 = \ln \left(\frac{v_1}{\sqrt{a^2+1}} + \sqrt{\frac{v_1^2}{a^2+1} + 1} \right)$$

$$t_2 = \ln \left(\frac{v_2}{\sqrt{a^2+1}} + \sqrt{\frac{v_2^2}{a^2+1} + 1} \right)$$

$$S = \frac{a^2+1}{2} \left[\frac{1}{2} \left(\left(\frac{v_2}{\sqrt{a^2+1}} + \sqrt{\frac{v_2^2}{a^2+1} + 1} \right)^2 - \left(\frac{v_1}{\sqrt{a^2+1}} + \sqrt{\frac{v_1^2}{a^2+1} + 1} \right)^2 \right) + \left(\sqrt{\frac{v_2^2}{a^2+1} + 1} - \frac{v_2}{\sqrt{a^2+1}} \right)^2 - \left(\sqrt{\frac{v_1^2}{a^2+1} + 1} - \frac{v_1}{\sqrt{a^2+1}} \right)^2 \right]$$

$$+ \ln \left[\left(\sqrt{\frac{v_2^2}{a^2+1} + 1} + \frac{v_2}{\sqrt{a^2+1}} \right) \left(\sqrt{\frac{v_1^2}{a^2+1} + 1} - \frac{v_1}{\sqrt{a^2+1}} \right) \right]$$

練習題 11.1

1. (1) 因為 $\lim_{x \rightarrow +\infty} \frac{x}{\frac{1}{x}} = 1$, 所以 $\int_1^{+\infty} \frac{x}{1+x^2} dx$ 發散

(2) 因為 $\lim_{x \rightarrow +\infty} \frac{\frac{3x^3-2}{x^5-x^3+1}}{\frac{1}{x^2}} = 3$, 所以 $\int_0^{+\infty} \frac{3x^3-2}{x^5-x^3+1} dx$ 收斂

(3) 因為 $\lim_{x \rightarrow +\infty} \frac{\frac{1}{x\sqrt{x^2+1}}}{\frac{1}{x^2}} = 1$, 所以 $\int_1^{+\infty} \frac{1}{x\sqrt{x^2+1}} dx$ 收斂

(4) 因為 $\int_e^{+\infty} \frac{1}{x(\log x)^p} dx = \int_1^{+\infty} \frac{1}{y^p} dy$, 所以當且僅當 $p > 1$ 時收斂

(5) 因為 $\int_1^{+\infty} x^{s-1} e^{-x} dx = \int_e^{+\infty} (\log y)^{s-1} \cdot \frac{1}{y^2} dy = \int_e^{+\infty} \frac{1}{y^2 (\log y)^{1-s}} dy$, 而 $\int_e^{+\infty} \frac{1}{y^p (\log y)^q} dy$ 收斂當且僅當 $p > 1$ 或 $(p=1 \text{ 且 } q > 1)$, 所以 $\forall s \in \mathbb{R}$, $\int_1^{+\infty} \frac{x^{s-1}}{e^x} dx$ 都收斂

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(6) 因為 $\lim_{x \rightarrow +\infty} \frac{(\log x)^p}{1+x^2} = 1$, 所以 $\forall p \in \mathbb{R}$, 有 $\int_1^{+\infty} \frac{(\log x)^p}{1+x^2} dx$ 收斂

2. 反證法：如果 $b \neq 0$,

因為 $\int_a^{+\infty} f(x) dx$ 收斂, 所以 $\forall \varepsilon > 0, \exists M > 0, \forall A > M$ 且 $p > 0$

$$\text{都有 } \left| \int_A^{A+p} f(x) dx \right| < \varepsilon$$

又知 $\lim_{x \rightarrow +\infty} f(x) = b > 0$, 取 $\varepsilon_0 = \frac{|b|}{2}$, $\exists M_0 > 0, \forall x > M_0$, 都有 $\frac{|b|}{2} < |f(x)| < \frac{3|b|}{2}$

因此, $\forall A > \max(M, M_0)$ 且 $p > 0$, 都有 $\varepsilon > \left| \int_A^{A+p} f(x) dx \right| = \int_A^{A+p} |f(x)| dx > \frac{|b|}{2}$,

與 ε 的任意性矛盾。由此可知, $b = 0$

$$5. \text{證: } \int_0^{+\infty} f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \int_0^1 f\left(\left(ax - \frac{b}{x}\right)^2\right) dx + \int_1^{+\infty} f\left(\left(ax - \frac{b}{x}\right)^2\right) dx$$

$$\text{設 } y = ax - \frac{b}{x} \Leftrightarrow xy = ax^2 - b \Leftrightarrow ax^2 - yx - b = 0$$

$$\Leftrightarrow x = \frac{y \pm \sqrt{y^2 + 4ab}}{2a}, \text{ 結合 } x \geq 0 \text{ 可知 } x = \frac{y + \sqrt{y^2 + 4ab}}{2a}$$

$$\Rightarrow dx = \frac{1}{2a} \left(1 + \frac{2y}{2\sqrt{y^2 + 4ab}}\right) dy = \frac{1}{2a} \left(1 + \frac{y}{\sqrt{y^2 + 4ab}}\right) dy$$

$$\text{所以 } \int_0^{+\infty} f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \int_{-\infty}^{a-b} f(y^2) \cdot \frac{1}{2a} \left(1 + \frac{y}{\sqrt{y^2 + 4ab}}\right) dy + \int_{a-b}^{+\infty} f(y^2) \cdot \frac{1}{2a} \left(1 + \frac{y}{\sqrt{y^2 + 4ab}}\right) dy$$

$$= \frac{1}{2a} \left[\int_{-\infty}^{+\infty} f(y^2) dy + \int_{-\infty}^{+\infty} f(y^2) \cdot \frac{y}{\sqrt{y^2 + 4ab}} dy \right]$$

偶函數

奇函數

$$= \frac{1}{a} \int_0^{+\infty} f(y^2) dy = \frac{1}{a} \int_0^{+\infty} f(x^2) dx$$

$$\text{取 } f(x) = e^{-x^2}, \text{ 則 } \int_0^{+\infty} \frac{e^{-\left(ax - \frac{b}{x}\right)^2}}{e^{-x^2}} dx = \frac{1}{a} \int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2a}$$

練習題 11.2

1. (1) 因為 $\forall A > 0$, $\left| \int_0^A \cos x \, dx \right| = |\sin A| \leq 1$, 記 $g(x) = \frac{\sqrt{x}}{1+x}$.

$$g'(x) = \frac{\frac{1}{2\sqrt{x}} - \sqrt{x}}{(1+x)^2} = \frac{1-x}{2\sqrt{x}(1+x)^2}. \text{ 當 } x > 1 \text{ 時, } g'(x) < 0$$

而且 $\lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{1+x} = 0$. 因此, 根據 Dirichlet 判別法知, $\int_0^{+\infty} \frac{\sqrt{x} \cos x}{1+x} \, dx$ 收斂

$$(3) \int_1^{+\infty} \frac{\cos^2 x}{x} \, dx = \frac{1}{2} \left(\int_1^{+\infty} \frac{\cos 2x}{x} \, dx + \int_1^{+\infty} \frac{1}{x} \, dx \right)$$

因為 $\left| \int_1^A \cos 2x \, dx \right| = \frac{1}{2} |\sin 2A - \sin 2| \leq 1$, 且 $\left(\frac{1}{x}\right)' = -\frac{1}{x^2} < 0$, $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$

所以 $\int_1^{+\infty} \frac{\cos 2x}{x} \, dx$ 收斂。但 $\int_1^{+\infty} \frac{1}{x} \, dx = +\infty$, 所以 $\int_1^{+\infty} \frac{\cos^2 x}{x} \, dx$ 發散

2. (2) $\left| \frac{\cos(1-2x)}{\sqrt{x^3} \sqrt[3]{x^2+1}} \right| \leq \frac{1}{x^{\frac{3}{2}}}$, 所以 $\int_1^{+\infty} \frac{\cos(1-2x)}{\sqrt{x^3} \sqrt[3]{x^2+1}} \, dx$ 絕對收斂

$$(4) \left| \frac{\sin x}{x \log x} \right| \geq \frac{\sin^2 x}{x \log x} = \frac{1}{2} \left(\frac{1}{x \log x} - \frac{\cos 2x}{x \log x} \right)$$

因為 $\left| \int_2^A \cos 2x \, dx \right| = \frac{1}{2} |\sin 2A - \sin 4| \leq 1$, 而且 $\frac{1}{x \log x}$ 單調遞減且趨向於 0

所以 $\int_2^{+\infty} \frac{\cos 2x}{x \log x} \, dx$ 收斂。但是 $\int_2^{+\infty} \frac{1}{x \log x} \, dx = \ln(\ln x) \Big|_2^{+\infty} = +\infty$

所以, $\int_2^{+\infty} \frac{\sin x}{x \log x} \, dx$ 不是絕對收斂。又因為 $\left| \int_2^A \sin x \, dx \right| \leq 2$

所以 $\int_2^{+\infty} \frac{\sin x}{x \log x} \, dx$ 條件收斂

3. 取 $b > \min(a, 0)$, 由題意知 $\int_b^{+\infty} f(x) \, dx$ 收斂, 加上 f 為非負減函數
由練習題 11.1 的第 2 題知, $\lim_{x \rightarrow +\infty} f(x) = 0$

因為 $\int_b^{+\infty} \frac{1}{x} \, dx = +\infty$, 所以 $\lim_{x \rightarrow +\infty} \frac{f(x)}{\frac{1}{x}} = 0 \Leftrightarrow f(x) = o\left(\frac{1}{x}\right)$

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4. 考慮積分 $\int_a^{+\infty} f(x) \cos 2x \, dx$ 。因為 $\left| \int_a^{+\infty} \cos 2x \, dx \right| = \frac{1}{2} |\sin 2A - \sin 2a| \leq 2$

而且 f 在 $[a, +\infty)$ 上遞減趨於 0，根據 Dirichlet 判別法知

$$\int_a^{+\infty} f(x) \cos 2x \, dx \text{ 收斂}$$

另一方面， $\int_a^{+\infty} f(x) \cos 2x \, dx = \int_a^{+\infty} f(x) \, dx - 2 \int_a^{+\infty} f(x) \sin^2 x \, dx$

因此， $\int_a^{+\infty} f(x) \, dx$ 與 $\int_a^{+\infty} f(x) \sin^2 x \, dx$ 同斂散。

問題 11.2

3. $\int_0^{+\infty} (-1)^{[x]} \, dx = \int_0^{+\infty} \frac{(-1)^{[t]}}{2\sqrt{t}} \, dt$ ，因為 $\frac{1}{\sqrt{t}}$ 單調趨向於 0，所以只需證明 $\forall A > 0$

$$\int_0^A (-1)^{[t]} \, dt \text{ 有界}$$

$$\left| \int_0^A (-1)^{[t]} \, dt \right| = \left| \int_0^{[A]} (-1)^{[t]} \, dt + \int_{[A]}^A (-1)^{[t]} \, dt \right|$$

注意到， $\forall n \in \mathbb{N}$ ， $\int_n^{n+1} (-1)^{[t]} \, dt = (-1)^{n+1}$ ， $\int_n^{n+2} (-1)^{[t]} \, dt = (-1)^{n+1} + (-1)^{n+2} = 0$

因此 $\left| \int_0^{[A]} (-1)^{[t]} \, dt \right| \leq 1$ ，即 $\left| \int_0^A (-1)^{[t]} \, dt \right| \leq 1 + \left| \int_{[A]}^A (-1)^{[t]} \, dt \right| \leq 1 + (-1)^{[A]} \{A\} < 2$

由 Dirichlet 判別法知， $\int_0^{+\infty} (-1)^{[x]} \, dx$ 收斂

6. $\int_a^{+\infty} f(x) \, dx = \lim_{x \rightarrow +\infty} f(x) - f(a) \Leftrightarrow \lim_{x \rightarrow +\infty} f(x) \text{ 存在且有限}$

因為 $\int_a^{+\infty} f(x) \, dx$ 收斂，所以根據練習題 11.1 第 2 題的結論（前面已證）

$$\lim_{x \rightarrow +\infty} f(x) = 0$$

練習題 11.3

1. (1) 當 $\beta = 0$ 時, 原積分 $= \int_0^{+\infty} \frac{x^a}{2} dx$, 當且僅當 $a < -1$ 時收斂

當 $\beta > 0$ 時 $\lim_{x \rightarrow +\infty} \frac{\frac{x^a}{1+x^\beta}}{\frac{x^a}{x^\beta}} = 1$, 當且僅當 $\beta - a > 1 \Leftrightarrow a < \beta - 1$ 時收斂

$$(2) \int_0^{+\infty} \frac{\log(1+x)}{x^a} dx = \int_0^1 \frac{\log(1+x)}{x^a} dx + \int_1^{+\infty} \frac{\log(1+x)}{x^a} dx$$

因為 $\forall x > 0$, $\frac{\log(1+x)}{x^a} > 0$, 所以上方兩個積分同斂散

$$\lim_{x \rightarrow +\infty} \frac{\frac{\log(1+x)}{x^a}}{\frac{\log(x)}{x^a}} = 1 \Rightarrow \int_1^{+\infty} \frac{\log(1+x)}{x^a} dx \text{ 與 } \int_1^{+\infty} \frac{1}{x^a (\log x)^a} dx \text{ 同斂散}$$

因此, 當且僅當 $a > 1$ 時, $\int_1^{+\infty} \frac{\log(1+x)}{x^a} dx$ 收斂

$$\text{另一方面, } \int_0^1 \frac{\log(1+x)}{x^a} dx = \int_{+\infty}^1 \frac{\log(1+\frac{1}{y})}{(\frac{1}{y})^a} (-\frac{1}{y^2}) dy = \int_1^{+\infty} \frac{\log(1+\frac{1}{y})}{y^{2-a}} dy$$

$$\lim_{y \rightarrow +\infty} \frac{\frac{\log(1+\frac{1}{y})}{y^{2-a}}}{\frac{1}{y^{3-a}}} = 1 \Rightarrow \int_1^{+\infty} \frac{\log(1+\frac{1}{y})}{y^{2-a}} dy \text{ 與 } \int_1^{+\infty} \frac{1}{y^{3-a}} dy \text{ 的斂散性一致}$$

即收斂條件是 $3-a > 1 \Leftrightarrow a < 2$

所以, 當且僅當 $1 < a < 2$ 時, 原積分收斂

$$(6) \int_1^{+\infty} \frac{1}{x^p (\log x)^q} dx = \int_1^2 \frac{1}{x^p (\log x)^q} dx + \int_2^{+\infty} \frac{1}{x^p (\log x)^q} dx$$

因為 $\forall x > 1$, $\frac{1}{x^p (\log x)^q} > 0$, 所以上面兩個積分同收斂

$$\int_2^{+\infty} \frac{dx}{x^p (\log x)^q} \text{ 收斂 } \Leftrightarrow p > 1 \text{ 或 } (p=1 \text{ 且 } q > 1)$$

$$\int_1^2 \frac{dx}{x^p (\log x)^q} = \int_{+\infty}^1 \frac{-\frac{1}{y^2} dy}{(1+\frac{1}{y})^p (\log(1+\frac{1}{y}))^q} = \int_1^{+\infty} \frac{1}{(1+\frac{1}{y})^p (\log(1+\frac{1}{y}))^q} dy$$

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因為 $\lim_{y \rightarrow +\infty} \frac{\frac{1}{y^2}}{\frac{(1+\frac{1}{y})^p (\log(1+\frac{1}{y}))^q}{1}} = 1$, 所以 $\int_1^{+\infty} \frac{1}{y^2} dy$ 與 $\int_1^{+\infty} \frac{1}{y^{2-q}} dy$ 同斂散

即當且僅當 $2-q > 1 \Leftrightarrow q < 1$ 時, $\int_1^{+\infty} \frac{dx}{x^p (\log x)^q}$ 斂

綜上所述, 當且僅當 $p > 1$ 且 $q < 1$ 時, 原積分斂

2. (1) $\int_0^{+\infty} x^p \sin(x^q) dx, q \neq 0$

解: $\int_0^{+\infty} x^p \sin(x^q) dx \xrightarrow{y=x^q} \int_L^u y^{\frac{p}{q}} \sin y \cdot \frac{1}{q} y^{\frac{1}{q}-1} dy = \frac{1}{q} \int_L^u y^{\frac{p+1}{q}-1} \sin y dy$

當 $q > 0$ 時, $L=0, u=+\infty$

當 $q < 0$ 時, $L=+\infty, u=0$

所以, 原積分 $= \frac{1}{|q|} \int_0^{+\infty} y^{\frac{p+1}{q}-1} \sin y dy = -\frac{1}{|q|} \left(\int_0^1 y^{\frac{p+1}{q}-1} \sin y dy + \int_1^{+\infty} y^{\frac{p+1}{q}-1} \sin y dy \right)$

$\lim_{y \rightarrow +\infty} \frac{y^{\frac{p+1}{q}-1} \sin y}{y^{\frac{p+1}{q}-1} y} = 1 \Leftrightarrow \int_0^1 y^{\frac{p+1}{q}-1} \sin y dy$ 與 $\int_0^1 y^{\frac{p+1}{q}-1} dy$ 的斂散性一致

其斂條件是 $\frac{p+1}{q} > -1$

$\int_1^{+\infty} \frac{\sin y}{y^{1-\frac{p+1}{q}}} dy$ 斂 $\Leftrightarrow \frac{p+1}{q} < 1$

綜上所述, 其斂條件是 $|\frac{p+1}{q}| < 1 \Leftrightarrow |p+1| < |q|$

因為 $\int_0^1 |y^{\frac{p+1}{q}-1} \sin y| dy = \int_0^1 y^{\frac{p+1}{q}-1} \sin y dy$, 所以只需考慮 $\int_1^{+\infty} y^{\frac{p+1}{q}-1} \sin y dy$

當且僅當 $1 - \frac{p+1}{q} > 1 \Leftrightarrow \frac{p+1}{q} < 0$ 時絕對斂

因此, 當 $\frac{p+1}{q} \in (-1, 0)$, 原積分絕對斂

當 $\frac{p+1}{q} \in [0, 1)$, 原積分條件斂

$$(2) \int_0^{+\infty} \frac{x^p \sin x}{1+x^8} dx, \quad \delta > 0$$

解: (i) 當 $\delta = 0$ 時, 原積分 $= \frac{1}{2} \int_0^{+\infty} x^p \sin x dx = \frac{1}{2} \left(\int_0^1 x^p \sin x dx + \int_1^{+\infty} x^p \sin x dx \right)$

因為 $\lim_{x \rightarrow 0^+} \frac{x^p \sin x}{x^{p+1}} = 1$, 所以 $\int_0^1 x^p \sin x dx$ 收斂 $\Leftrightarrow p > -2$

而 $\int_1^{+\infty} x^p \sin x dx$ 收斂 $\Leftrightarrow p < 0$

所以當 $\delta = 0$ 時, 收斂條件是 $-2 < p < 0$

因為 $|x^p \sin x| = x^p \sin x \quad (\forall x \in (0, 1))$, 所以只需考慮 $\int_1^{+\infty} x^p \sin x dx$

而 $\int_1^{+\infty} x^p \sin x dx$ 絕對收斂 $\Leftrightarrow p < -1$

因此, 當 $-2 < p < -1$ 時, 原積分絕對收斂

當 $-1 \leq p < 0$ 時, 原積分條件收斂

(ii) 當 $\delta > 0$ 時, $\lim_{x \rightarrow 0^+} \frac{x^p \sin x}{1+x^8} = 1 \Rightarrow \int_0^1 \frac{x^p \sin x}{1+x^8} dx$ 的收斂條件和絕對收斂條件都是 $p > -2$

$$\varepsilon > \left| \int_{-\pi}^{(R+\pi)x} \frac{x^p \sin x}{1+x^8} dx \right| = \left| \frac{\xi^p}{1+\xi^8} \right| \cdot 2 \Rightarrow p < 9$$

$$\begin{aligned} \text{同時 } \left(\frac{x^p}{1+x^8} \right)' &= \frac{px^{p-1}(1+x^8) - x^p \cdot 8x^7}{(1+x^8)^2} = \frac{px^{p-1} - (8-p)x^{p+8-1}}{(1+x^8)^2} \\ &= \frac{x^{p-1}}{(1+x^8)^2} \cdot [p - (8-p)x^8] < 0 \quad \left(\text{當 } x^8 > \frac{|p|}{8-p} \text{ 時} \right) \end{aligned}$$

因為 $\left| \int_A^B \sin x dx \right| \leq 2$, 所以根據 Dirichlet 判別法知

$$\int_1^{+\infty} \frac{x^p \sin x}{1+x^8} dx \text{ 收斂} \Rightarrow \text{綜上所述, 當且僅當 } -2 < p < 8 \text{ 時, 原積分收斂}$$

另一方面 $\left| \frac{x^p \sin x}{1+x^8} \right| \leq \frac{x^p}{1+x^8} \leq \frac{1}{x^{8-p}} \Rightarrow \text{當 } -2 < p < 8-1 \text{ 時, 原積分絕對收斂}$

$$\left| \frac{x^p \sin x}{1+x^8} \right| \geq \frac{1}{2} \left(\frac{x^p}{1+x^8} - \frac{x^p \cos 2x}{1+x^8} \right) \Rightarrow \text{當 } 8-1 \leq p < 8 \text{ 時, 原積分條件收斂}$$

6.

(i) $f \in C^{(1)}[0, +\infty)$, 且 $f(+\infty)$ 存在, $a, b > 0$

$$\text{證明: } \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(+\infty)) \log \frac{b}{a}$$

證: 首先考慮 $k, l > 0$

$$\begin{aligned} \int_k^l \frac{f(ax) - f(bx)}{x} dx &= \int_k^l \frac{f(ax)}{x} dx - \int_k^l \frac{f(bx)}{x} dx \\ &= \int_{ak}^{al} \frac{f(x)}{x} dx - \int_{bk}^{bl} \frac{f(x)}{x} dx \\ &= \int_{bk}^{al} \frac{f(x)}{x} dx - \int_{bk}^{ak} \frac{f(x)}{x} dx \\ &= \int_b^a \frac{f(lx)}{x} dx - \int_b^a \frac{f(kx)}{x} dx \\ &= \int_b^a \frac{f(lx) - f(kx)}{x} dx \end{aligned}$$

由積分中值定理知

$$\begin{aligned} \int_b^a \frac{f(lx) - f(kx)}{x} dx &= (f(l\xi) - f(k\xi)) \int_b^a \frac{1}{x} dx \\ &= (f(l\xi) - f(k\xi)) \log \frac{a}{b} \\ &= (f(k\xi) - f(l\xi)) \log \frac{b}{a}, \text{ 其中 } \xi = a + \theta(b-a), \theta \in [0, 1] \end{aligned}$$

因為 ξ 有界, 所以 $\lim_{k \rightarrow 0^+} k\xi = 0$ 。再根據 f 在 $[0, +\infty)$ 上的連續性知, $\lim_{k \rightarrow 0^+} f(k\xi) = f(0)$

$$\text{所以, } \int_0^l \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(l\xi)) \log \frac{b}{a}$$

另一方面, 因為 ξ 有正下界, 所以 $\lim_{l \rightarrow +\infty} l\xi = +\infty$ 。同理可得 $\lim_{l \rightarrow +\infty} f(l\xi) = f(+\infty)$

$$\text{所以 } \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(+\infty)) \log \frac{b}{a}$$

(2) $\int_1^{+\infty} \frac{f(x)}{x} dx$ 收敛 $\Leftrightarrow \forall \varepsilon > 0, \exists M > 1, \forall A, B > M$, 都有 $\left| \int_A^B \frac{f(x)}{x} dx \right| < \varepsilon$

$$\left| \int_A^B \frac{f(x)}{x} dx \right| = \left| f(\xi) \log \frac{B}{A} \right| < \varepsilon$$

$$\text{取 } B = eA \Rightarrow |f(\xi)| < \varepsilon$$

$$\text{令 } A \rightarrow +\infty, \varepsilon \rightarrow 0, \lim_{\xi \rightarrow +\infty} f(\xi) = 0$$

$$\text{所以 } \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = (f(0) - 0) \cdot \log \frac{b}{a} = f(0) \log \frac{b}{a}$$

$$(3) \int_0^1 \frac{f(x)}{x} dx = \int_{+\infty}^1 \frac{y f(\frac{1}{y})}{y^2} dy = \int_1^{+\infty} \frac{f(\frac{1}{y})}{y} dy$$

$$\text{同理可得 } \lim_{y \rightarrow +\infty} f(\frac{1}{y}) = 0 \Leftrightarrow \lim_{x \rightarrow 0^+} f(x) = 0, \text{ 即 } f(0) = 0$$

$$\text{因此 } \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = -f(+\infty) \log \frac{b}{a} = f(+\infty) \log \frac{a}{b}$$

7. (1) 记 $f(x) = e^{-x}$, $f(0) = 1$, $f(+\infty) = 0$

$$\Rightarrow \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} dx = \log \frac{b}{a}$$

(2) 记 $f(x) = \cos x$, $f(0) = 0$, $\boxed{\int_1^{+\infty} \frac{\cos x}{x} dx}$ 由 Dirichlet 判别法知, 此积分收敛

$$\Rightarrow \int_0^{+\infty} \frac{\cos ax - \cos bx}{x} dx = 0$$